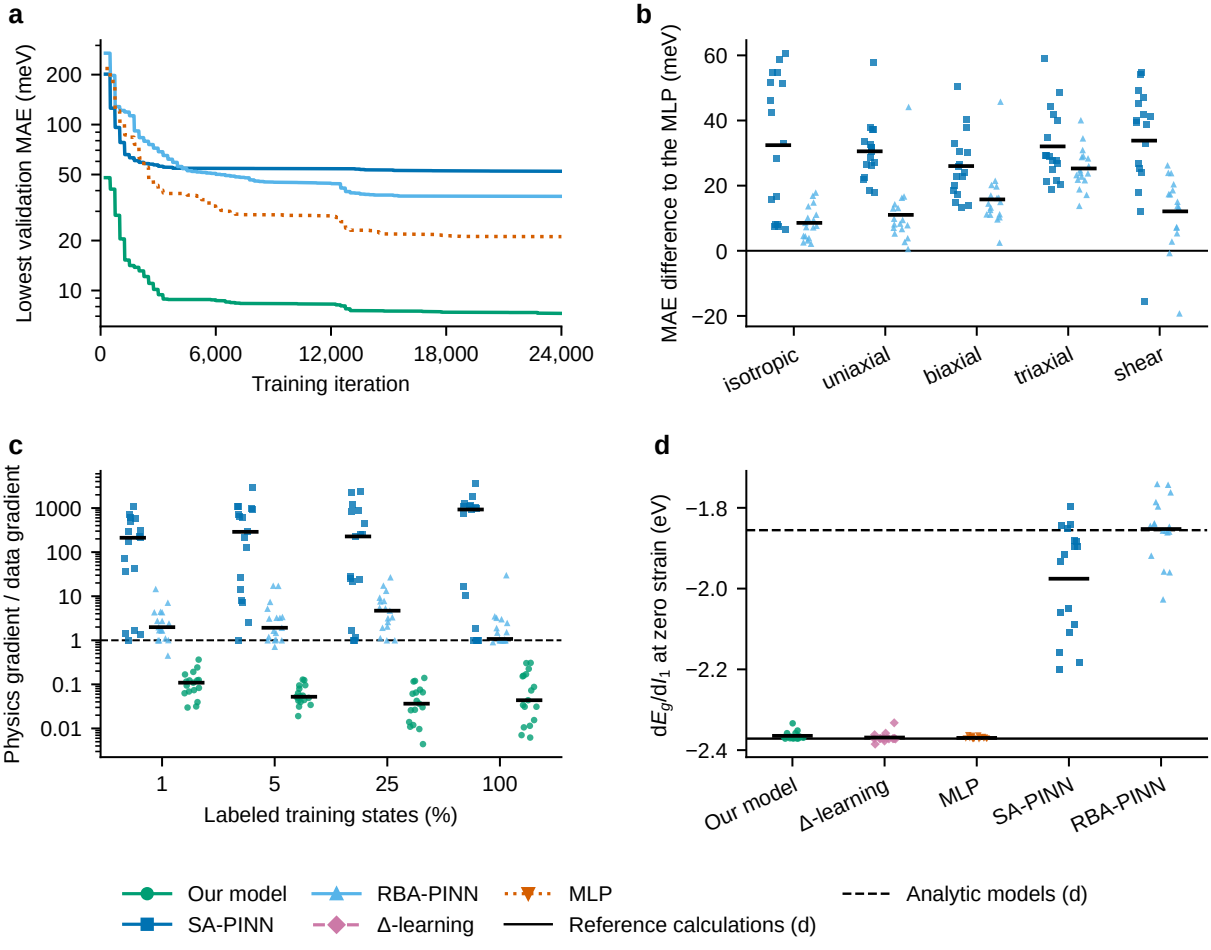
a
Lowest validation MAE (meV)
200
100
50
20
10
0
6,000
12,000
18,000
24,000
Training iteration
b
MAE difference to the MLP (meV)
60
40
20
0
−20
isotropic
uniaxial
biaxial
triaxial
shear
c
Physics gradient / data gradient
1000
100
10
1
0.1
0.01
1
5
25
100
Labeled training states (%)
d
dE_g/dl_1 at zero strain (eV)
−1.8
−2.0
−2.2
−2.4
Our model
Δ-learning
MLP
SA-PINN
RBA-PINN
Our model
SA-PINN
RBA-PINN
Δ-learning
MLP
Reference calculations (d)
Analytic models (d)